# Supporting Information

SIG-Net: Similarity-Informed Local-Competence Mixtures of Cross-Fitted Experts for Molecular Property Prediction under Leakage-Audited Out-of-Distribution Splits

## S1. Pre-specified analysis plan

The analysis plan below was written after the final SIG-Net run was launched and before any of its test-set results were inspected. The SIG-Net design was fixed using validation scores on three development folds. The plan and its two addenda are reproduced with their original wording and timestamps; only the formatting has been converted from Markdown.

### Pre-specified analysis plan for SIG-Net v3

Written 2026-10-17 ~23:35 local time, after the v3 RunPod run was launched and **before any v3 test-set result was viewed**. The v3 design (experts, cross-fitting K=5, grid for k/tau/gamma) was fixed using validation scores on three development folds (FreeSolv bm_scaffold s42, ESOL bm_scaffold s42, ESOL butina_cluster s42); no test scores were inspected during design.

#### Data

- 4 datasets x 4 split protocols x 5 seeds = 80 folds; identical splits/fold preparation for v2 and v3 runs.
- v2 results (`results/folds/`) and v3 results (`results/folds_v3/`) are merged per fold.

#### Primary endpoint

- SIG-Net v3 (Proposed) vs. the strongest non-v3 comparator, defined as the non-v3 model with the best average rank across the 16 settings. Test: two-sided Wilcoxon signed-rank test on per-fold relative improvement over all 80 folds. Also reported: number of settings where v3 has the better 5-seed mean.

#### Secondary endpoints

1. v3 vs. "Global stack (same cross-fitted experts)" — isolates the contribution of local competence + novelty shrinkage.
2. v3 vs. each bagged expert and each v2-run baseline (same Wilcoxon procedure, 80 folds).
3. Average rank over all models (Friedman test, Nemenyi CD).
4. Error stratified by nearest-neighbour Tanimoto similarity (OOD folds, regression datasets).

#### Reporting rules

- All 16 settings reported, including those where v3 is not best.
- No post-hoc removal of datasets, splits, seeds or baselines.
- FreeSolv generic_scaffold has a single distinct test set across seeds; it is reported but flagged.
- If v3 does not beat the strongest comparator on the primary endpoint, this is stated as the result.

#### Addendum (2026-10-18 ~00:55, written before any BBBP or ChemBERTa-2 result was viewed)

Following external review of the draft, two extensions were added. The SIG-Net design (experts, K, grids) is unchanged.

1. **BBBP** (MoleculeNet blood-brain barrier penetration, classification) is added as a fifth dataset, curated and split with the identical Phase 1 code (4 protocols x 5 seeds). All models, including SIG-Net, are run on its 20 folds.
2. **Pretrained foundation-model baselines:** frozen ChemBERTa-2 (DeepChem/ChemBERTa-77M-MTR) mean-pooled embeddings with (a) a ridge/logistic head and (b) an XGBoost head, both selected on the validation fold, on all 100 folds. ChemBERTa-2 is a comparison model only; it is not added as a SIG-Net expert (that would change the design post hoc).

Reporting: the main text reports all 5 datasets (100 folds). The pre-specified 80-fold primary endpoint (4 original datasets, original comparator set) is reported unchanged alongside it. No other analysis changes.

**Addendum 2 — SIG-Net v4 study (2026-10-18 ~09:40, before any confirmatory result was computed)**

**Motivation:** reviewer request to simplify or justify the novelty exponent and to strengthen the local router.

**Splits:** fresh seeds generated with the identical splitting code. Development seed **11** (all 20 settings) may be used for design decisions, including its test folds. Confirmatory seeds **1, 2, 3** (60 folds) are not inspected until the design is frozen. FreeSolv generic_scaffold yields the same test set for every seed, so it is **excluded from the confirmatory endpoint** (reported separately).

**Experts:** 9 cross-fitted experts (the 7 of v3 + ChemBERTa-2/XGBoost + XGBoost/ECFP4); all OOF/val/test matrices saved.

**Candidate routers (development):** v4 (local error-covariance routing with novelty shrinkage toward the global covariance, composite ECFP/MACCS/physchem kernel, novelty-damped local residual correction), on 7 or 9 experts; v3 router; global stacks; and v4 ablations.

**Design-selection rule (fixed now, before the remaining development folds arrived):** the v4 variant with the highest mean per-fold relative improvement over the 7-expert global stack across the 20 seed-11 folds is chosen (ties: more wins).

**Confirmatory primary endpoint:** chosen v4 vs the 7-expert global stack over the 56 confirmatory folds (60 minus FreeSolv generic), two-sided Wilcoxon signed-rank test on per-fold relative improvement. **Secondary:** v4 vs v3 router; v4 vs 9-expert global stack; v4 ablations (no residual correction, no composite kernel, no novelty shrinkage).

**Reporting:** whatever the confirmatory result is, it is reported; the v3 paper results are unchanged.

**Design freeze (2026-10-18 09:54, before any confirmatory result was computed)** Applying the rule above to the 20 development folds (file results/dev_v4_seed11_FROZEN.csv): **v4 full (9 experts)** has the highest mean improvement over the 7-expert global stack (+0.68%, median +0.51%, 14/20 wins) and is frozen as **SIG-Net v4**: 9 experts; grid k in {10,25,50}, a in {0.5,0.75,1.0}, gamma in {0,0.5,1,2,4,global}, nu in {0,0.25,0.5,0.75}, gamma_d in {1,2}, rho = 0.05; all selected per fold on the validation fold. Ablations are those listed above.

# S2. Full benchmark

Table S1: Mean test metric over five seeds for every model. (a) Regression tasks, test RMSE (lower is better). R: random, BM: Bemis–Murcko scaffold, G: generic skeleton, Bu: Butina cluster.

| Model | Free-R | Free-BM | Free-G | Free-Bu | ESOL-R | ESOL-BM | ESOL-G | ESOL-Bu | Lipo-R | Lipo-BM | Lipo-G | Lipo-Bu |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| Anchor (Multi-View Stack) [5-fold bagged] | 0.964 | 1.870 | 1.430 | 1.083 | 0.545 | 0.767 | 0.753 | 0.577 | 0.594 | 0.687 | 0.688 | 0.791 |
| Anchor-Only (Multi-View Stack) | 0.939 | 1.870 | 1.430 | 1.083 | 0.545 | 0.767 | 0.753 | 0.577 | 0.595 | 0.687 | 0.688 | 0.791 |
| AttentiveFP | 0.983 | 2.030 | 1.363 | 1.097 | 0.570 | 0.849 | 0.807 | 0.602 | 0.545 | 0.614 | 0.634 | 0.724 |
| AttentiveFP [5-fold bagged] | 0.898 | 1.910 | 1.348 | 1.167 | 0.585 | 0.810 | 0.771 | 0.614 | 0.528 | 0.584 | 0.605 | 0.687 |
| ChemBERTa-2 (77M-MTR) + XGBoost | 2.009 | 3.026 | 2.569 | 1.786 | 0.847 | 1.128 | 0.994 | 0.839 | 0.743 | 0.873 | 0.860 | 0.964 |
| ChemBERTa-2 (77M-MTR) + linear | 1.386 | 2.309 | 1.809 | 1.397 | 0.793 | 1.038 | 1.289 | 0.839 | 0.768 | 0.892 | 0.842 | 0.976 |
| D-MPNN (Chemprop) | 1.111 | 1.823 | 1.337 | 1.219 | 0.652 | 0.861 | 0.890 | 0.641 | 0.527 | 0.646 | 0.636 | 0.717 |
| D-MPNN + Deep CORAL | 0.871 | 1.714 | 1.526 | 1.079 | 0.556 | 0.775 | 0.743 | 0.567 | 0.547 | 0.660 | 0.671 | 0.766 |
| D-MPNN + Deep CORAL [5-fold bagged] | 0.878 | 1.821 | 1.558 | 1.135 | 0.533 | 0.756 | 0.728 | 0.552 | 0.541 | 0.651 | 0.645 | 0.739 |
| D-MPNN + RDKit-2D | 0.871 | 1.723 | 1.525 | 1.089 | 0.557 | 0.777 | 0.736 | 0.568 | 0.544 | 0.665 | 0.670 | 0.764 |
| D-MPNN + RDKit-2D Ensemble (3x) | 0.873 | 1.759 | 1.537 | 1.107 | 0.536 | 0.773 | 0.748 | 0.560 | 0.540 | 0.653 | 0.660 | 0.747 |
| D-MPNN + RDKit-2D [5-fold bagged] | 0.870 | 1.795 | 1.556 | 1.115 | 0.536 | 0.762 | 0.730 | 0.565 | 0.542 | 0.648 | 0.649 | 0.744 |
| Dual-Stream Gated Residual (v1 design) | 0.980 | 1.792 | 1.432 | 1.114 | 0.542 | 0.799 | 0.760 | 0.566 | 0.571 | 0.680 | 0.665 | 0.791 |
| GATv2 | 1.234 | 1.976 | 1.395 | 1.346 | 0.567 | 0.861 | 0.803 | 0.624 | 0.574 | 0.628 | 0.662 | 0.763 |
| GCN | 1.011 | 2.160 | 1.279 | 1.237 | 0.583 | 0.853 | 0.742 | 0.607 | 0.616 | 0.683 | 0.706 | 0.788 |
| GINE | 0.996 | 2.014 | 1.335 | 1.216 | 0.560 | 0.774 | 0.756 | 0.579 | 0.556 | 0.628 | 0.638 | 0.714 |
| GINE [5-fold bagged] | 0.978 | 1.996 | 1.381 | 1.238 | 0.563 | 0.819 | 0.740 | 0.593 | 0.557 | 0.610 | 0.632 | 0.722 |
| Global stack (same cross-fitted experts) | 0.822 | 1.719 | 1.465 | 1.081 | 0.516 | 0.763 | 0.696 | 0.544 | 0.506 | 0.576 | 0.590 | 0.670 |
| HighwayMLP (ECFP4 + RDKit-2D) | 0.971 | 1.812 | 1.565 | 1.064 | 0.561 | 0.821 | 0.771 | 0.609 | 0.575 | 0.655 | 0.679 | 0.745 |
| RF (ECFP4-2048) | 2.309 | 3.765 | 2.818 | 2.261 | 1.241 | 1.654 | 1.639 | 1.295 | 0.886 | 0.982 | 0.930 | 1.032 |
| RF (MACCS-167) | 1.141 | 1.905 | 1.690 | 1.389 | 1.058 | 1.334 | 1.359 | 1.002 | 0.800 | 0.918 | 0.896 | 0.958 |
| RF (RDKit-2D) | 1.148 | 2.349 | 1.878 | 1.324 | 0.623 | 0.835 | 0.832 | 0.645 | 0.668 | 0.765 | 0.771 | 0.879 |
| RF (RDKit-2D) [5-fold bagged] | 1.283 | 2.452 | 1.842 | 1.379 | 0.641 | 0.840 | 0.816 | 0.653 | 0.675 | 0.767 | 0.772 | 0.880 |
| SIG-Net v2 (Proposed) | 0.868 | 1.753 | 1.415 | 1.074 | 0.515 | 0.751 | 0.734 | 0.549 | 0.530 | 0.636 | 0.633 | 0.730 |
| SIG-Net v2 w/o ensemble | 0.878 | 1.743 | 1.397 | 1.065 | 0.523 | 0.746 | 0.722 | 0.554 | 0.530 | 0.641 | 0.635 | 0.733 |
| SIG-Net v3 (Proposed) | 0.800 | 1.720 | 1.405 | 1.038 | 0.511 | 0.754 | 0.704 | 0.544 | 0.505 | 0.580 | 0.589 | 0.668 |
| SIG-Net v3 w/o novelty shrinkage | 0.797 | 1.687 | 1.415 | 1.038 | 0.512 | 0.741 | 0.703 | 0.544 | 0.509 | 0.592 | 0.596 | 0.675 |
| Stacked (Ensemble + Anchor, global w) | 0.875 | 1.783 | 1.427 | 1.094 | 0.515 | 0.752 | 0.734 | 0.550 | 0.533 | 0.636 | 0.634 | 0.730 |
| Stacked (XGBoost + D-MPNN) | 0.884 | 1.880 | 1.474 | 1.057 | 0.532 | 0.751 | 0.717 | 0.563 | 0.531 | 0.637 | 0.636 | 0.735 |
| XGBoost (ECFP4 + RDKit-2D) | 1.066 | 2.464 | 1.773 | 1.157 | 0.597 | 0.788 | 0.787 | 0.594 | 0.597 | 0.693 | 0.698 | 0.809 |
| XGBoost (ECFP4 + RDKit-2D) [5-fold bagged] | 1.152 | 2.309 | 1.755 | 1.195 | 0.592 | 0.765 | 0.763 | 0.605 | 0.607 | 0.694 | 0.701 | 0.808 |
| XGBoost (ECFP4-2048) | 1.697 | 3.229 | 2.437 | 1.771 | 1.126 | 1.530 | 1.464 | 1.210 | 0.794 | 0.874 | 0.866 | 0.978 |
| XGBoost (RDKit-2D) | 0.998 | 2.251 | 1.585 | 1.099 | 0.596 | 0.792 | 0.787 | 0.604 | 0.609 | 0.700 | 0.715 | 0.823 |

Table S1: (continued) (b) Classification tasks, test ROC-AUC (higher is better).

| Model | BACE-R | BACE-BM | BACE-G | BACE-Bu | BBBP-R | BBBP-BM | BBBP-G | BBBP-Bu |
|---|---|---|---|---|---|---|---|---|
| Anchor (Multi-View Stack) [5-fold bagged] | 0.901 | 0.855 | 0.798 | 0.763 | 0.931 | 0.937 | 0.950 | 0.887 |
| Anchor-Only (Multi-View Stack) | 0.902 | 0.855 | 0.798 | 0.763 | 0.931 | 0.937 | 0.950 | 0.887 |
| AttentiveFP | 0.872 | 0.836 | 0.806 | 0.704 | 0.892 | 0.898 | 0.923 | 0.818 |
| AttentiveFP [5-fold bagged] | 0.881 | 0.856 | 0.781 | 0.772 | 0.901 | 0.901 | 0.932 | 0.834 |

| Model | BACE-R | BACE-BM | BACE-G | BACE-Bu | BBBP-R | BBBP-BM | BBBP-G | BBBP-Bu |
|---|---|---|---|---|---|---|---|---|
| ChemBERTa-2 (77M-MTR) + XGBoost | 0.868 | 0.838 | 0.819 | 0.800 | 0.907 | 0.914 | 0.936 | 0.858 |
| ChemBERTa-2 (77M-MTR) + linear | 0.885 | 0.840 | 0.799 | 0.730 | 0.876 | 0.895 | 0.932 | 0.841 |
| D-MPNN (Chemprop) | 0.887 | 0.840 | 0.731 | 0.745 | 0.906 | 0.929 | 0.912 | 0.821 |
| D-MPNN + Deep CORAL | 0.879 | 0.832 | 0.807 | 0.780 | 0.925 | 0.914 | 0.931 | 0.856 |
| D-MPNN + Deep CORAL [5-fold bagged] | 0.881 | 0.843 | 0.812 | 0.795 | 0.926 | 0.915 | 0.935 | 0.855 |
| D-MPNN + RDKit-2D | 0.879 | 0.830 | 0.812 | 0.781 | 0.926 | 0.916 | 0.927 | 0.856 |
| D-MPNN + RDKit-2D Ensemble (3x) | 0.879 | 0.837 | 0.813 | 0.782 | 0.925 | 0.921 | 0.927 | 0.853 |
| D-MPNN + RDKit-2D [5-fold bagged] | 0.883 | 0.840 | 0.816 | 0.782 | 0.926 | 0.914 | 0.934 | 0.848 |
| Dual-Stream Gated Residual (v1 design) | 0.900 | 0.855 | 0.799 | 0.758 | 0.932 | 0.939 | 0.944 | 0.879 |
| GATv2 | 0.853 | 0.823 | 0.701 | 0.681 | 0.892 | 0.902 | 0.906 | 0.823 |
| GCN | 0.813 | 0.776 | 0.689 | 0.714 | 0.926 | 0.915 | 0.923 | 0.846 |
| GINE | 0.764 | 0.794 | 0.703 | 0.615 | 0.912 | 0.918 | 0.924 | 0.822 |
| GINE [5-fold bagged] | 0.862 | 0.831 | 0.716 | 0.720 | 0.916 | 0.921 | 0.938 | 0.844 |
| Global stack (same cross-fitted experts) | 0.901 | 0.863 | 0.810 | 0.767 | 0.935 | 0.939 | 0.952 | 0.881 |
| HighwayMLP (ECFP4 + RDKit-2D) | 0.888 | 0.852 | 0.782 | 0.767 | 0.914 | 0.917 | 0.932 | 0.841 |
| RF (ECFP4-2048) | 0.893 | 0.844 | 0.783 | 0.780 | 0.918 | 0.904 | 0.934 | 0.834 |
| RF (MACCS-167) | 0.880 | 0.830 | 0.772 | 0.734 | 0.926 | 0.930 | 0.943 | 0.888 |
| RF (RDKit-2D) | 0.880 | 0.846 | 0.790 | 0.793 | 0.925 | 0.932 | 0.948 | 0.874 |
| RF (RDKit-2D) [5-fold bagged] | 0.883 | 0.847 | 0.794 | 0.790 | 0.927 | 0.932 | 0.949 | 0.871 |
| SIG-Net v2 (Proposed) | 0.901 | 0.853 | 0.818 | 0.765 | 0.933 | 0.940 | 0.944 | 0.888 |
| SIG-Net v2 w/o ensemble | 0.902 | 0.854 | 0.817 | 0.765 | 0.934 | 0.938 | 0.946 | 0.888 |
| SIG-Net v3 (Proposed) | 0.900 | 0.864 | 0.815 | 0.790 | 0.940 | 0.940 | 0.954 | 0.887 |
| SIG-Net v3 w/o novelty shrinkage | 0.901 | 0.863 | 0.818 | 0.790 | 0.939 | 0.940 | 0.954 | 0.886 |
| Stacked (Ensemble + Anchor, global w) | 0.899 | 0.848 | 0.816 | 0.760 | 0.934 | 0.938 | 0.941 | 0.876 |
| Stacked (XGBoost + D-MPNN) | 0.892 | 0.841 | 0.807 | 0.762 | 0.931 | 0.935 | 0.936 | 0.856 |
| XGBoost (ECFP4 + RDKit-2D) | 0.889 | 0.849 | 0.789 | 0.759 | 0.926 | 0.933 | 0.949 | 0.855 |
| XGBoost (ECFP4 + RDKit-2D) [5-fold bagged] | 0.891 | 0.849 | 0.789 | 0.773 | 0.926 | 0.935 | 0.950 | 0.862 |
| XGBoost (ECFP4-2048) | 0.899 | 0.844 | 0.779 | 0.705 | 0.912 | 0.896 | 0.919 | 0.813 |
| XGBoost (RDKit-2D) | 0.882 | 0.843 | 0.777 | 0.742 | 0.924 | 0.928 | 0.949 | 0.864 |

## S3. Full rankings

Table S2: Mean rank over 20 settings for all models except internal SIG-Net ablations.

| # | Model | Mean rank | OOD mean rank | First | Top-3 |
|---|---|---|---|---|---|
| 1 | SIG-Net | 2.60 | 2.87 | 12 | 14 |
| 2 | Two-expert novelty gate | 5.20 | 5.80 | 1 | 7 |
| 3 | Two-expert gate (single D-MPNN) | 5.50 | 5.93 | 3 | 6 |
| 4 | Stack: D-MPNN ens. + anchor | 7.50 | 8.20 | 0 | 3 |
| 5 | Stack: XGBoost + D-MPNN | 9.65 | 10.27 | 0 | 3 |
| 6 | Multi-view anchor (5-fold bagged) | 11.38 | 11.73 | 0 | 2 |
| 7 | Multi-view anchor | 11.62 | 12.07 | 0 | 1 |
| 8 | D-MPNN + Deep CORAL [5-fold bagged] | 11.65 | 12.00 | 0 | 1 |
| 9 | D-MPNN + RDKit-2D [5-fold bagged] | 11.90 | 12.73 | 0 | 1 |
| 10 | Dual-stream gated residual | 12.25 | 12.87 | 0 | 1 |
| 11 | D-MPNN + RDKit-2D Ensemble (3x) | 13.35 | 13.53 | 0 | 0 |
| 12 | D-MPNN + RDKit-2D | 14.20 | 14.27 | 0 | 1 |
| 13 | D-MPNN + Deep CORAL | 14.30 | 14.33 | 1 | 1 |
| 14 | AttentiveFP [5-fold bagged] | 14.85 | 14.33 | 0 | 5 |
| 15 | HighwayMLP (ECFP4 + RDKit-2D) | 16.40 | 16.33 | 0 | 1 |
| 16 | GINE [5-fold bagged] | 16.55 | 15.67 | 0 | 2 |
| 17 | XGBoost (ECFP4 + RDKit-2D) [5-fold bagged] | 16.70 | 16.13 | 0 | 1 |
| 18 | XGBoost (ECFP4 + RDKit-2D) | 18.05 | 18.00 | 0 | 0 |
| 19 | GINE | 18.30 | 17.60 | 0 | 2 |
| 20 | D-MPNN (Chemprop) | 18.45 | 18.53 | 0 | 2 |
| 21 | AttentiveFP | 18.50 | 17.80 | 0 | 0 |
| 22 | XGBoost (RDKit-2D) | 19.10 | 18.67 | 0 | 0 |
| 23 | RF (RDKit-2D) [5-fold bagged] | 19.70 | 19.40 | 0 | 0 |
| 24 | RF (RDKit-2D) | 19.90 | 19.07 | 0 | 1 |
| 25 | GCN | 21.10 | 21.13 | 1 | 1 |
| 26 | GATv2 | 22.30 | 21.87 | 0 | 0 |
| 27 | RF (MACCS-167) | 23.10 | 22.93 | 0 | 1 |
| 28 | ChemBERTa-2 (77M-MTR) + XGBoost | 23.50 | 22.13 | 2 | 2 |
| 29 | ChemBERTa-2 (77M-MTR) + linear | 25.55 | 25.47 | 0 | 0 |

## S4. All paired comparisons (100 folds)

Table S3: SIG-Net vs. every other model. Relative improvement per fold (%), two-sided Wilcoxon signed-rank test.

| Comparator | Folds won | Settings won | Mean | Median | $p$ |
|---|---|---|---|---|---|
| SIG-Net w/o novelty shrinkage | 27/100 | 9/20 | +0.05 | +0.00 | $p = 0.166$ |
| Global stack (same experts) | 60/100 | 15/20 | +0.80 | +0.14 | $p = 0.0035$ |
| Two-expert gate (single D-MPNN) | 68/100 | 14/20 | +2.63 | +1.58 | $p = 1.6 \times 10^{-5}$ |
| Two-expert novelty gate | 69/100 | 15/20 | +2.67 | +1.47 | $p = 8.2 \times 10^{-6}$ |
| Stack: D-MPNN ens. + anchor | 73/100 | 19/20 | +3.18 | +1.89 | $p = 2.0 \times 10^{-7}$ |
| Stack: XGBoost + D-MPNN | 80/100 | 19/20 | +3.85 | +2.75 | $p = 1.3 \times 10^{-9}$ |
| D-MPNN + Deep CORAL [5-fold bagged] | 86/100 | 19/20 | +4.57 | +3.67 | $p = 2.0 \times 10^{-13}$ |

| | | | | | |
|---|---|---|---|---|---|
| D-MPNN + RDKit-2D [5-fold bagged] | 88/100 | 19/20 | +4.72 | +4.38 | $p = 1.6 \times 10^{-13}$ |
| D-MPNN + RDKit-2D Ensemble (3x) | 83/100 | 20/20 | +4.87 | +4.50 | $p = 1.7 \times 10^{-11}$ |
| D-MPNN + Deep CORAL | 82/100 | 19/20 | +5.18 | +4.64 | $p = 3.8 \times 10^{-11}$ |
| D-MPNN + RDKit-2D | 84/100 | 19/20 | +5.23 | +4.72 | $p = 2.4 \times 10^{-11}$ |
| AttentiveFP [5-fold bagged] | 87/100 | 19/20 | +5.31 | +3.87 | $p = 9.2 \times 10^{-12}$ |
| Dual-stream gated residual | 80/100 | 19/20 | +5.45 | +2.56 | $p = 2.4 \times 10^{-11}$ |
| Multi-view anchor | 76/100 | 18/20 | +5.70 | +2.48 | $p = 6.2 \times 10^{-10}$ |
| Multi-view anchor (5-fold bagged) | 76/100 | 18/20 | +5.83 | +2.59 | $p = 3.3 \times 10^{-10}$ |
| HighwayMLP (ECFP4 + RDKit-2D) | 87/100 | 20/20 | +6.95 | +5.93 | $p = 3.0 \times 10^{-12}$ |
| AttentiveFP | 87/100 | 19/20 | +7.41 | +6.32 | $p = 1.7 \times 10^{-13}$ |
| GINE [5-fold bagged] | 90/100 | 19/20 | +7.53 | +5.87 | $p = 6.3 \times 10^{-13}$ |
| D-MPNN (Chemprop) | 89/100 | 19/20 | +8.96 | +6.42 | $p = 8.7 \times 10^{-15}$ |
| XGBoost (RDKit-2D) | 91/100 | 20/20 | +9.28 | +8.44 | $p = 6.5 \times 10^{-16}$ |
| GINE | 85/100 | 19/20 | +9.35 | +6.09 | $p = 3.3 \times 10^{-11}$ |
| XGBoost (ECFP4 + RDKit-2D) | 90/100 | 20/20 | +9.75 | +7.36 | $p = 1.7 \times 10^{-14}$ |
| XGBoost (ECFP4 + RDKit-2D) [5-fold bagged] | 89/100 | 20/20 | +9.78 | +6.82 | $p = 7.8 \times 10^{-15}$ |
| GCN | 87/100 | 19/20 | +10.66 | +9.04 | $p = 1.6 \times 10^{-12}$ |
| GATv2 | 92/100 | 19/20 | +11.40 | +9.10 | $p = 4.5 \times 10^{-15}$ |
| RF (RDKit-2D) | 91/100 | 20/20 | +13.40 | +13.59 | $p = 5.3 \times 10^{-16}$ |
| RF (RDKit-2D) [5-fold bagged] | 93/100 | 20/20 | +14.10 | +13.68 | $p = 5.2 \times 10^{-17}$ |
| ChemBERTa-2 (77M-MTR) + linear | 95/100 | 20/20 | +20.77 | +23.06 | $p = 3.1 \times 10^{-17}$ |
| RF (MACCS-167) | 95/100 | 19/20 | +21.46 | +20.65 | $p = 3.3 \times 10^{-16}$ |
| ChemBERTa-2 (77M-MTR) + XGBoost | 93/100 | 18/20 | +23.41 | +29.94 | $p = 2.6 \times 10^{-16}$ |
| XGBoost (ECFP4-2048) | 96/100 | 20/20 | +28.27 | +32.56 | $p = 1.0 \times 10^{-17}$ |
| RF (ECFP4-2048) | 93/100 | 20/20 | +31.43 | +38.16 | $p = 1.5 \times 10^{-16}$ |

## S5. Confirmation on fresh splits: all comparisons

Table S4: SIG-Net v4 vs. every router and cross-fitted expert on the confirmatory folds (seeds 1–3).

| Comparator | Folds won | Mean (%) | Median (%) | $p$ |
|---|---|---|---|---|
| Global stack (7) [primary] | 35/57 | +1.27 | +0.20 | $p = 0.020$ |
| SIG-Net v3 router (7) | 34/57 | +0.91 | +0.19 | $p = 0.117$ |
| Global stack (9) | 34/57 | +1.34 | +0.21 | $p = 0.019$ |
| v4 w/o residual correction | 29/57 | +0.29 | +0.01 | $p = 0.016$ |
| v4 w/o composite kernel | 17/57 | +0.38 | +0.00 | $p = 0.857$ |
| v4 w/o novelty shrinkage | 22/57 | +0.02 | +0.00 | $p = 0.633$ |
| D-MPNN + RDKit-2D [5-fold bagged] | 48/57 | +5.10 | +2.43 | $p = 7.3 \times 10^{-7}$ |
| D-MPNN + Deep CORAL [5-fold bagged] | 46/57 | +5.54 | +3.78 | $p = 4.1 \times 10^{-7}$ |
| GINE [5-fold bagged] | 45/57 | +5.70 | +4.76 | $p = 1.9 \times 10^{-5}$ |
| AttentiveFP [5-fold bagged] | 47/57 | +5.09 | +2.67 | $p = 5.5 \times 10^{-7}$ |
| XGBoost (ECFP4 + RDKit-2D) [5-fold bagged] | 50/57 | +10.10 | +5.04 | $p = 2.1 \times 10^{-9}$ |
| RF (RDKit-2D) [5-fold bagged] | 48/57 | +13.91 | +12.02 | $p = 2.4 \times 10^{-8}$ |
| Anchor (Multi-View Stack) [5-fold bagged] | 42/57 | +6.18 | +3.06 | $p = 1.2 \times 10^{-5}$ |
| ChemBERTa-2 + XGBoost [5-fold bagged] | 55/57 | +22.87 | +25.19 | $p = 2.1 \times 10^{-10}$ |
| XGBoost (ECFP4) [5-fold bagged] | 54/57 | +28.21 | +32.41 | $p = 2.0 \times 10^{-10}$ |
| Original router vs Global stack (7) | 28/57 | +0.27 | +0.00 | $p = 0.805$ |
| Global stack (7), FreeSolv generic included | 38/60 | +1.64 | +0.26 | $p = 0.0052$ |

## S6. Per-setting results

Table S5: SIG-Net vs. the best other external model in each setting (mean ± SD over 5 seeds); paired $t$-test over seeds (descriptive; low power with $n = 5$).

| Dataset | Split | SIG-Net | Best other | Model | Rank | Seed wins | $p$ |
|---|---|---|---|---|---|---|---|
| BACE | BM scaffold | 0.864 ± 0.046 | 0.856 ± 0.054 | AttentiveFP [5-fold bagged] | 1 | 4/5 | 0.326 |
| BACE | Butina cluster | 0.790 ± 0.076 | 0.800 ± 0.060 | ChemBERTa-2 (77M-MTR) + XGBoost | 5 | 3/5 | 0.735 |
| BACE | Generic skeleton | 0.815 ± 0.081 | 0.819 ± 0.048 | ChemBERTa-2 (77M-MTR) + XGBoost | 6 | 3/5 | 0.811 |
| BACE | Random | 0.900 ± 0.021 | 0.902 ± 0.021 | Two-expert gate (single D-MPNN) | 5 | 2/5 | 0.441 |
| BBBP | BM scaffold | 0.940 ± 0.029 | 0.940 ± 0.030 | Two-expert novelty gate | 2 | 3/5 | 0.807 |
| BBBP | Butina cluster | 0.887 ± 0.033 | 0.888 ± 0.024 | Two-expert gate (single D-MPNN) | 6 | 3/5 | 0.844 |
| BBBP | Generic skeleton | 0.954 ± 0.025 | 0.950 ± 0.017 | XGBoost (ECFP4 + RDKit-2D) [5-fold bagged] | 1 | 4/5 | 0.527 |
| BBBP | Random | 0.940 ± 0.007 | 0.934 ± 0.011 | Two-expert gate (single D-MPNN) | 1 | 3/5 | 0.386 |
| ESOL | BM scaffold | 0.754 ± 0.112 | 0.746 ± 0.072 | Two-expert gate (single D-MPNN) | 5 | 3/5 | 0.737 |
| ESOL | Butina cluster | 0.544 ± 0.080 | 0.549 ± 0.084 | Two-expert novelty gate | 1 | 3/5 | 0.681 |
| ESOL | Generic skeleton | 0.704 ± 0.093 | 0.717 ± 0.096 | Stack: XGBoost + D-MPNN | 1 | 2/5 | 0.339 |
| ESOL | Random | 0.511 ± 0.042 | 0.515 ± 0.024 | Stack: D-MPNN ens. + anchor | 1 | 3/5 | 0.746 |
| FreeSolv | BM scaffold | 1.720 ± 0.136 | 1.714 ± 0.176 | D-MPNN + Deep CORAL | 2 | 2/5 | 0.950 |
| FreeSolv | Butina cluster | 1.038 ± 0.155 | 1.057 ± 0.133 | Stack: XGBoost + D-MPNN | 1 | 3/5 | 0.668 |
| FreeSolv | Generic skeleton | 1.405 ± 0.054 | 1.279 ± 0.058 | GCN | 9 | 0/5 | 0.027 |
| FreeSolv | Random | 0.800 ± 0.211 | 0.868 ± 0.168 | Two-expert novelty gate | 1 | 4/5 | 0.174 |
| Lipophilicity | BM scaffold | 0.580 ± 0.037 | 0.584 ± 0.039 | AttentiveFP [5-fold bagged] | 1 | 4/5 | 0.556 |
| Lipophilicity | Butina cluster | 0.668 ± 0.041 | 0.687 ± 0.057 | AttentiveFP [5-fold bagged] | 1 | 3/5 | 0.405 |
| Lipophilicity | Generic skeleton | 0.589 ± 0.033 | 0.605 ± 0.033 | AttentiveFP [5-fold bagged] | 1 | 5/5 | 0.017 |
| Lipophilicity | Random | 0.505 ± 0.047 | 0.527 ± 0.044 | D-MPNN (Chemprop) | 1 | 4/5 | 0.067 |